

Unit :- 12

Unit's name :- Basic Concept of Organic chemistry

Organic compound

- Those compounds which contain carbon, hydrogen and its derivatives is called organic compound. For example :- Methane, Formic acid, etc.

Inorganic Compound

- Those compounds except carbon, hydrogen and its derivatives is called inorganic compound. For example: NaCl, HCl, $MgSO_4$, $CaCO_3$, etc.

Organic Chemistry

- A branch of chemistry that deals with the study of organic compound is called organic chemistry.

Vital Force Theory

- In 1815, Jons Jacob Berzelius suggested that an organic compound is prepared by living organism through mysterious force called vital force. This theory is also known as vitalism.

Note :- In 1828, Friedrich Wohler was prepared an organic compound in lab i.e. Urea (NH_2CONH_2)

Reasons for the separate study for organic compound from inorganic compound :-

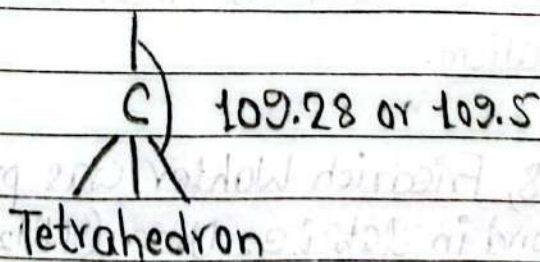
- 1- An organic compound contains long-chain of carbon.
- 2- An organic compound having own distinguish properties.

Difference between organic compound and inorganic compound

Organic Compound	Inorganic Compound
i) Mostly covalent compound.	i) Generally ionic compound but few are covalent compound.
ii) They are generally insoluble in water but soluble in organic solvent. For eg: alcohol, ether, acetone, etc.	ii) They are soluble in water but insoluble in organic solvent.
iii) They have low melting and boiling points.	iii) They have high melting and boiling points.
iv) They are highly combustible.	iv) They are generally non-combustible.
v) They show isomerism.	v) They don't show isomerism

Tetravalency

A carbon has capacity to combine four other atoms to form covalent bond. Such a tendency is called tetravalency. If carbon bonded with another four atoms, it gives tetrahedron structure with bond angle 109.28 or 109.5

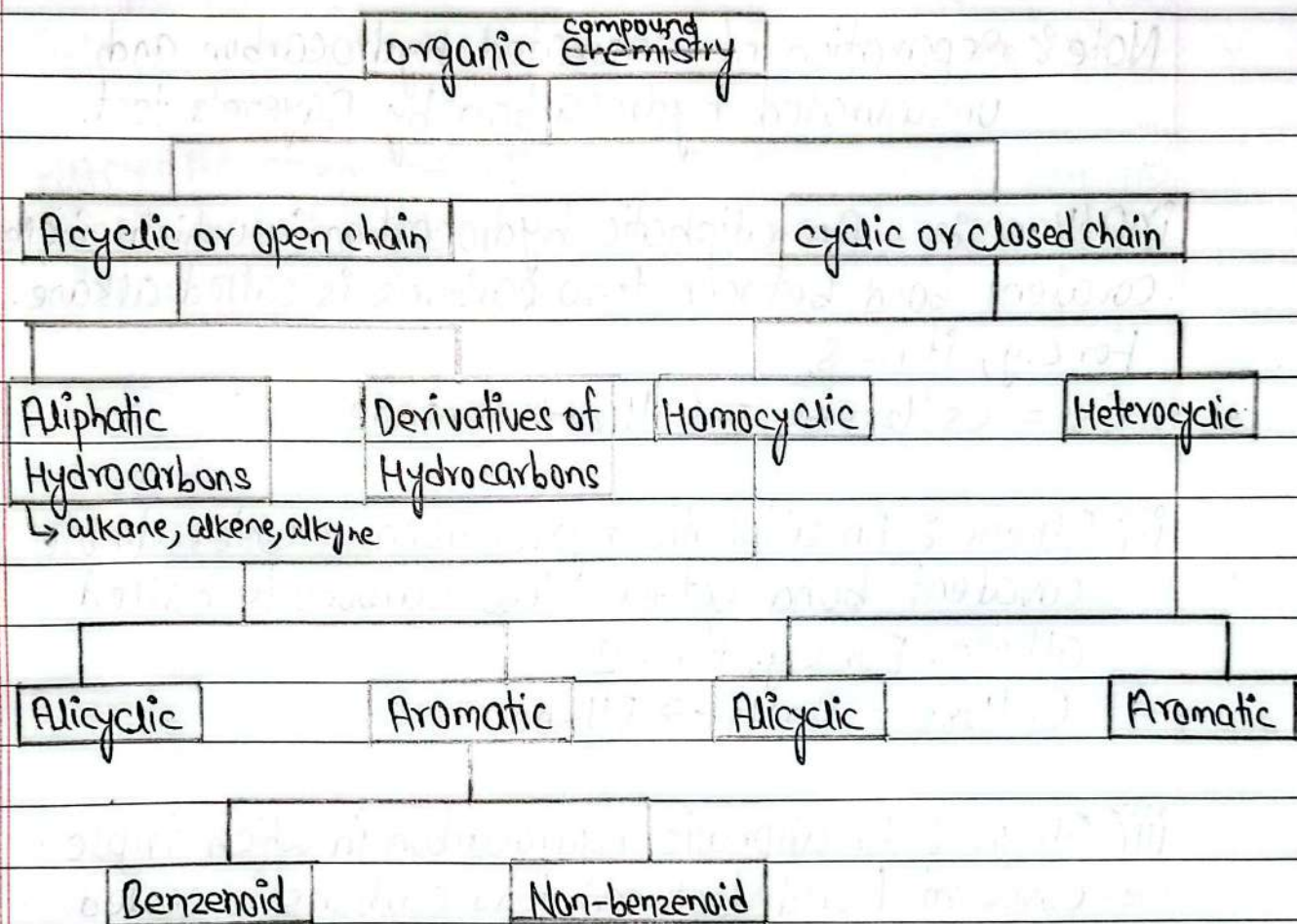


Catenation

A carbon has a special property to form covalent bond

with carbon giving straight chain, branch chain and ring.
Such a property is called catenation.

classification of organic compound



- 1) Acyclic compound or open chain.
An organic compound which have non-cyclic or open chain of carbon is called acyclic compound. They are further divided into :-

- a) Aliphatic Hydrocarbon
b) Derivatives of hydrocarbon

- a) Aliphatic Hydrocarbon :- An organic compound which have open chain of carbon and is made of only

Carbon and hydrogen is called aliphatic hydrocarbon. They are also classified into three types :-

i) Alkane (C_nH_{2n+2})

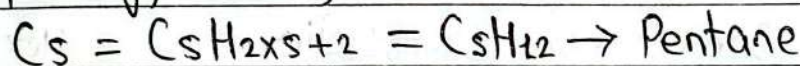
ii) Alkene (C_nH_{2n})

iii) Alkyne (C_nH_{2n-2})

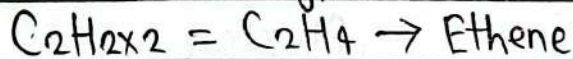
Note: Separation of saturated hydrocarbon and unsaturated hydrocarbon by Bayere's test.

i) Alkane: An aliphatic hydrocarbon in which single covalent bond between two carbons is called alkane.

For e.g., if $n=5$,

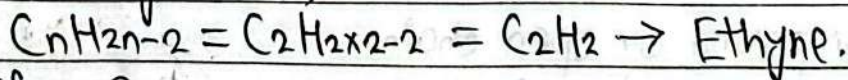


ii) Alkene: An aliphatic hydrocarbon in which double covalent bond between two carbons is called alkene. For e.g., if $n=2$

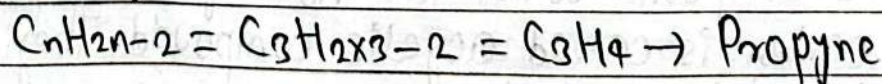


iii) Alkyne: An aliphatic hydrocarbon in which triple covalent bond between two carbons is called alkyne.

For e.g., $n=2$,



If $n=3$,



b) Derivates of hydrocarbon: An organic compound in which one or more than one hydrogen replaced by functional group is called derivatives of hydrocarbon. For e.g. ethanal, ethanol, etc.

27 cyclic compound

An organic compound in which are cyclic or ring or closed form is called cyclic compound. They are two types:

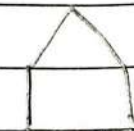
Homo cyclic compound (Carbocyclic compound)

It is defined as if only carbon forms ring in organic compound. They are further classified into two types:

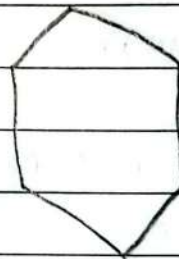
i) Alicyclic compound

ii) Aromatic compound.

i) Alicyclic compound: An organic compound which are cyclic chain form but have properties similar to the open chain is called alicyclic compound. For e.g.,



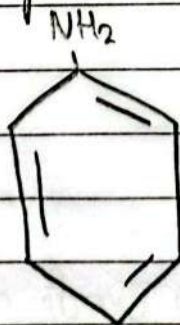
cyclopentane



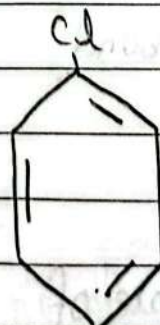
cyclohexane.

ii) Aromatic compound: An organic compound which behaviour like a benzene is called aromatic compound.

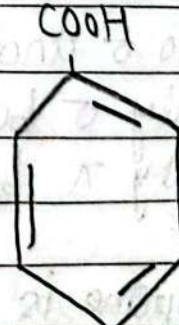
For e.g.:



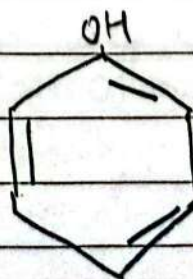
aniline



Chlorobenzene



benzoic acid



Phenol

MCQs

1. The vital force theory was proposed by
 - a. Dalton
 - b. Wohler
 - c. Berzelius ✓
 - d. Avogadro

2. The shape of methane is.
 - a. tetrahedral ✓
 - b. Octahedral
 - c. linear
 - d. Square planar

3. An organic compound of aromatic compound burns with air
 - a. sooty flame. ✓
 - b. non-sooty flame
 - c. both (a) and (b)
 - d. None

4. Acetylene consist of _____
 - a. one σ and two π bond ✓
 - b. two σ and one π bond
 - c. only σ bond
 - d. only π bond

5. Propane is an example of
 - a. Straight chain ✓ (connected in a continuous chain $\rightarrow$ $\text{CH}_3\text{-CH}_2\text{-CH}_3$)
 - b. branched chain
 - c. cyclic compound ✓
 - d. None

6. Which of the following statement is correct for an organic compound?

- a. Have low M.P.
- b. Are covalent compound
- c. soluble in organic solvent.
- d. All of these ✓

7. Which of the following organic compound is used an anaesthetic purpose? during surgery?

- a. ether
- b. chloroform ✓ (trichloromethane) $\rightarrow \text{CHCl}_3$
- c. methanol
- d. both a and b

8. Alkane is

- a. Paraffin ✓ (ie. inert active)
- b. olefin
- c. acetylene
- d. None

9. Ethylene is

- a. C_2H_6
- b. C_2H_4 ✓
- c. C_2H_2
- d. CH_4

10. Carbon forms a large no. of compounds because

- a. is tetravalency
- b. has variable valency
- c. has the property of catenation. ✓
- d. has great chemical affinity.

11. Oldest known organic compound is

- a. CH_3COOH ✓ (It occurs naturally from the fermentation of ethanol)
- b. NH_2CONH_2
- c. $\text{CH}_3\text{CH}_2\text{OH}$
- d. CH_3NH_2

12. An unsaturated aliphatic compound is

- a. C_2H_6
- b. C_2H_2 ✓
- c. C_3H_8
- d. CH_4

13. Carbon word came from

- a. Greek
- b. Latin ✓ ("carbo" means 'coal' or 'charcoal')
- c. French
- d. English

14. Which of the following is ~~not~~ known as king of elements?

- a. Gold
- b. Hydrogen
- c. Carbon ✓ (due to its unique ability to form stable compounds)
- d. Nitrogen

15. Fat dissolve in

- a. ether
- b. alcohol ✓
- c. water
- d. benzene (Fat is non-polar and benzene is non-polar)

16. The universal organic solvent is

- a. petrol

- b. water
- c. ether — (mostly non-polar but has slight polarity)
- d. alcohol
17. Which of the following has low M.P.?
- a. vegetable ghee. — (made of fats/oils with weak van der Waals forces)
- b. sodium bicarbonate
- c. calcium chloride
- d. Rock salt
18. Which of the following is not an example of organic compound?
- a. CHCl_3
- b. CH_3COOH
- c. CCl_4
- d. None —
19. Bond angle in methane is
- a. 109.5° —
- b. 109.6°
- c. 109.7°
- d. 109.8°
20. Tetravalency concept was proposed by
- a. Berzelius
- b. Wohler
- c. Van't Hoff and Le Bel —
- d. Dalton.

The metals occurring in the form of their compound in nature are called

- a. Minerals ✓
- b. Flux
- c. Matrix
- d. Ore

A mineral is known as an ore if the metal

- a. cannot be produced from it.
- b. cannot be produced from it.
- c. can be produced from it profitably ✓
- d. is very costly

Flux is a substance that used for

- a. Reducing metal oxide.
- b. Removing silica
- c. Removing unwanted metal oxide. ✓
- d. none of these

The impurities present in a mineral is called.

- a. Flux
- b. alloy
- c. Gangue ✓
- d. slag

Iron ore is concentrated by

- a. Froth floatation
- b. Electrolysin
- c. Roasting
- d. Magnetic separation. ✓

Froth floatation process is used for the metallurgy of

- chlorides ores.
- oxides ores
- Sulphide ore. ✓
- Carbonate ores.

The main function of roasting is

- oxidation
- reduction ✓
- to remove volatile matter
- to make slag

Slag = Flux + Gangue

Ques.-1. Write the difference between calcination and roasting.

Ques.-2. Definition of flux and its type.

Ques.-3. Definition of matrix

Ques.-4. Definition of slag

Ques.-5. Write a difference between minerals and ores.

Answers.

Calcination		Roasting	
1.	It is the process where metallic ore is heated below its melting point and in absence of air, hence the product is obtained in metallic oxide.	1.	It is the process where metallic ore is converted to metallic oxide in presence of air and is heated below its boiling point.
2.	It is especially done for oxide, hydroxides and carbonate ores.	2.	It is usually done for sulphide ores.

Flux and its type

→ Flux is the substance added to the ores before heating, with the object that it will react chemically with the impurities (gangue) present in the ores and form a fusible slag.

There are 3 types of fluxes:

1. **Acidic fluxes** : E.g., silica (SiO_2), borax ($\text{Na}_2\text{B}_4\text{O}_7$), P_2O_5 etc.
2. **Basic fluxes** : E.g., limestone (CaCO_3), magnesite (MgCO_3), haematite (Fe_2O_3), etc.
3. **Neutral fluxes** : E.g., fluor-spar (CaF_2), etc. which are generally used to maintain the fluidity of melted substances.

Definition of matrix/gangue

→ Ores contains a large amount of rocky materials as silicates, sand, mica, carbonates etc as impurities which is called as gangue or matrix (from Latin: matte-mother i.e. earthy or stony substance in which minerals are found embedded.)

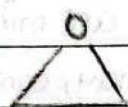
Definition of slag (Flux + gangue → slag)

→ Slags are the fusible substances which are obtained by the combination of flux with gangue and are easily removed in metallurgical process, because fused slag and molten metal form immiscible and separate layers at the bottom of the furnace and can be separated easily. Most slags are silicates formed by the combination of sand with base like FeO , CaO , Al_2O_3 , MnO , etc.

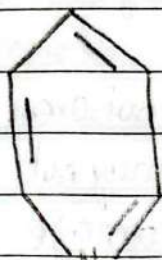
Differences Between Minerals and Ores

Minerals		Ores	
1.	Minerals are the natural inorganic that exists in earth crust.	1.	Ores are the Minerals from which metal can be extracted economically and conveniently.
2.	They have definite crystalline structure.	2.	They do not have definite crystalline structure.
3.	All minerals are not ores.	3.	All ores are minerals.
4.	Minerals may or may not contain a good percentage of metals.	4.	Ores always contains a good Percentage of metal.

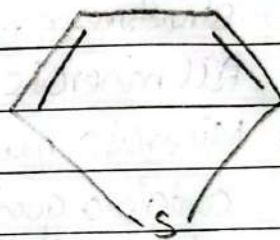
Heterocyclic Compound :- An organic compound in which at least one hetero atoms like oxygen, Nitrogen, sulphur, halogens form cyclic structure is called Heterocyclic compound. For e.g.,



Oxirane

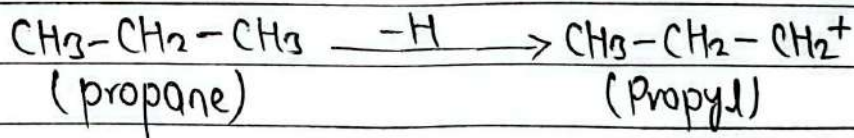
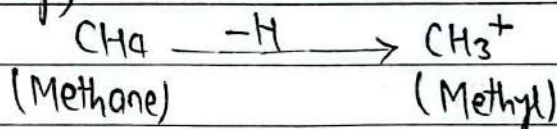


pyridine



Thiophene

Alkyl group (R-) :- If one hydrogen is removed from alkane then radical is called alkyl group.
For eg.,



Functional group :- It is atom or group of molecules present in an organic compound that determines the physical and chemical properties of that compound.

Functional group	Family name	General formula
-OH	Alcohol	R-OH
- $\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{H}$ or -CHO	Aldehyde	R-CHO or R- $\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{H}$
- $\overset{\overset{\text{O}}{\parallel}}{\text{C}}-$ or, -CO-	Ketone	R- $\overset{\overset{\text{O}}{\parallel}}{\text{C}}=\text{R}$ or R-CO-R
- $\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{O}-\text{R}$ or -COOR	ester	RCOOR
-O-	ether	ROR
-COOH or - $\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{O}-\text{H}$	Carboxylic acid	R-COOH
-CONH ₂	acid amide	R-CONH ₂
-NH ₂	amine	R-NH ₂
- $\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{O}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-$ or -COOCO-	acid anhydride	R- $\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{O}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{R}$
-CN	Cyanide(nitrile)	R-CN
-NC	isocyanide	R-NC
-SO ₃ H	Sulphonic acid	R-SO ₃ H
-X (Cl, Br, I, F)	halo compound	R-X
- $\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{Cl}$ or COCl	acid chloride	RCOCl

Homologous Series : A series in which an organic compound is arranged in the order of increasing molecular mass. belonging to same family or same functional group. is called Homologous series.

For e.g., Alkane ($\text{C}_n\text{H}_{2n+2}$)

If $n=1$, CH_4 = Methane

$n=2$, $\text{C}_2\text{H}_6 \rightarrow$ Ethane
(CH_3CH_3)

$n=3$, $\text{C}_3\text{H}_8 \rightarrow$ Propane
($\text{CH}_3\text{CH}_2\text{CH}_3$)

$n=4$, $\text{C}_4\text{H}_{10} \rightarrow$ Butane
($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$)

Alcohol ($C_nH_{2n+1}OH$) ; suffix $\rightarrow$ ol

$n=1$, $CH_3OH \rightarrow$ Methanol (Methyl alcohol)

$n=2$, $CH_3CH_2OH \rightarrow$ Ethanol (ethyl alcohol)

$n=3$, $CH_3CH_2CH_2OH \rightarrow$ Propanol (propyl alcohol)

$n=4$, $CH_3CH_2CH_2CH_2OH \rightarrow$ butanol (butyl alcohol)

Carboxylic acid ($C_nH_{2n+1}-COOH$)

$n=1$, $HCOOH$ - Methanoic acid (formic acid)

$n=2$, CH_3COOH - Ethanoic acid (acetic acid)

$n=3$, CH_3CH_2COOH - propanoic acid

$n=4$, $CH_3CH_2CH_2COOH$ - Butanoic acid

Characteristics of Homologous Series

- i) They have same general formula.
- ii) They have same chemical properties but difference in physical properties with increasing the atomic mass.
- iii) They have same general method of preparation.
- iv) Each member differs by $-CH_2$ unit or 14 amu.

Types of formula

Molecular formula :- It is a formula that indicates/represents the no. of atoms present in a molecule.

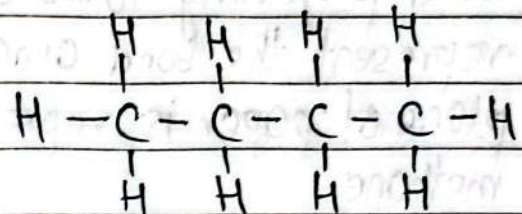
For e.g., In ethane, there are two carbon and six hydrogen atoms. So, its molecular formula is C_2H_6 .

Empirical formula :- It is a ratio of atoms present in a molecule. For e.g., In benzene, there are six carbon and six hydrogen atoms. So, its empirical formula is CH .

Empirical formula of Butane (C_4H_{10}) is C_2H_5 .

Structural formula (display formula) :- The formula that indicates how the atoms are bonded in a molecule is called structural formula.

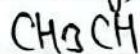
For e.g.,



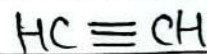
Butane

Condensed formula (contracted formula) :- The formula which is written by contracting or condensing hydrogen with carbon is called condensed formula. It is also defined as a condensed form of structural formula. In such formula, single bond and lone pair are omitted.

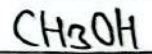
For e.g.,



Ethane



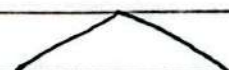
Ethyne



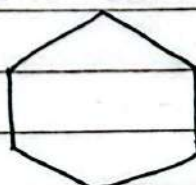
Methanol

Bond line formula (skeletal formula) :- The carbon and hydrogen of organic compounds are not shown in this type of formula. Each corner represents the carbon. Carbon to carbon bonds are represented by bond line notation. The hydrogen atoms are understood to satisfy the valency of carbon. Unsubstituted terminal corner represents methyl group.

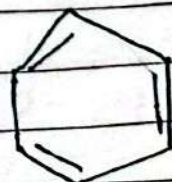
For e.g.,



Propane



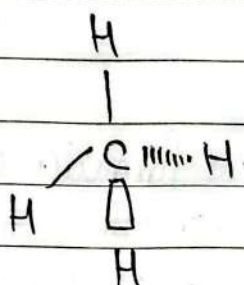
cyclohexane



Benzene

Solid wedge line notation formula (A 3D formula) :

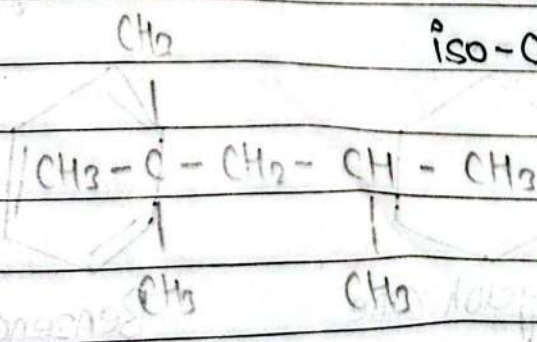
The three dimensional (3-D) structure of organic compound can be represented in planar paper by using solid (—) and dashed (.....) wedge formula. Solid wedge represent a bond projecting towards observer while dashed wedge represent the bond away from the observer. The bond lying on plane of paper is represented by a normal line (—). For methane,

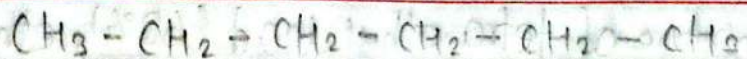


Methane

Octane No. : The fraction of crude oil having carbon no. C_6 - C_{10} is called gasoline/petrol. When compressed air and fuel is ignited in a cylinder, it gives energy. If quality of fuel is not good, it produce metallic sound is called Knocking property. Thus, the quality of fuel is measured in the term of anti-knocking properties and expressed in the term of Octane no.

Octane no. is defined as the percentage of iso octane in a mixture of iso octane and n heptane. The IUPAC name of iso-octane is 2,2,4-trimethyl Pentane





n heptane

Octane no. can be improved by

a) cracking

b) Isomerism

c) Alkylation

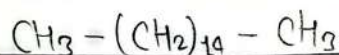
d) Adding additive

Octane no. is increasing order :

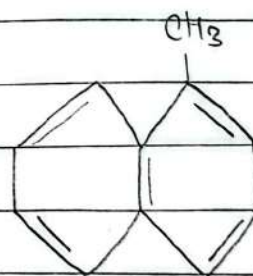
straight chain < branched < double or triple < cyclic < Aromatic

Cetane Number : It is the percentage of cetane present in a mixture of cetane and α -methyl naphthalene in diesel.

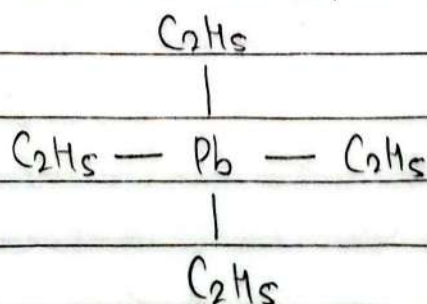
Cetane (n-hexadecane)



α -methyl naphthalene



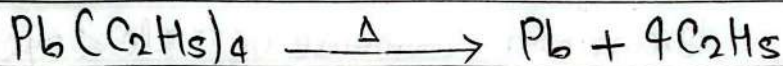
Gasoline additives : Those chemical substance which is added in petrol or diesel to minimize the knocking property is called gasoline additives. for e.g., TEL (tetraethyl lead)



TEL is the mixture of tetra ethyl lead (63%), ethylene bromide (26%), ethylene chloride (9%) and methylene blue (2%). The petrol in which TEL is added is called leaded petrol or ethyl petrol.

Working Mechanism of TEL.

TEL decomposed to produce free radical which combine with another radical produced by irregular combustion. And it forms large carbon chain to give regular combustion.



Lead is deposit in the cylinder which is removed by ethylene bromide as a lead bromide.

